

```

TITLE 'BACKGROUND MUSIC';
DATA BANK;
INPUT DAY $ TIME WEEK MUSIC $ PRODIND @@;
LABEL PRODIND='PRODUCTIVITY INDEX';
CARDS;
M  2 1 D 18  M  5 2 E 17  M  3 3 C 13  M  4 4 A 13  M  1 5 B 21
T  3 1 E 17  T  1 2 C 13  T  4 3 B 32  T  5 4 D 31  T  2 5 A  7
W  1 1 A 14  W  4 2 D 15  W  2 3 E 27  W  3 4 B 25  W  5 5 C 31
TH 5 1 B 21  TH 3 2 A 16  TH 1 3 D 29  TH 2 4 C 24  TH 4 5 E 21
F  4 1 C 17  F  2 2 B 34  F  5 3 A  7  F  1 4 E 21  F  3 5 D 26
;
ODS GRAPHICS ON;
PROC GLM PLOTS=(DIAGNOSTICS);
CLASS MUSIC DAY TIME WEEK;
MODEL PRODIND=MUSIC DAY TIME WEEK;
MEANS MUSIC/TUKEY LINES;
OUTPUT OUT=NEW P=PREDY R=RESID;
PROC UNIVARIATE NORMAL;
VAR RESID;
RUN;

```